Constraints on Late-Time $f\sigma_8$ Suppression from $\mu < 1$, $\Sigma = 1$: Planck 2018 and Large-Scale Structure

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Abstract

We investigate a scale-independent, late-time suppression of linear structure growth within the standard μ–Σ modified gravity framework. The model imposes $\mu(a) < 1$ for matter perturbations while preserving $\Sigma(a) = 1$, leaving photon geodesics and the CMB acoustic structure unmodified. The time dependence is governed by an activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ that vanishes at early times and reaches unity at $a = 1$, preserving recombination-era physics by construction. For the theoretically motivated coupling $\beta = \Omega_m/2$, the model yields a present-day gravitational coupling $\mu(z{=}0) = 0.865$, corresponding to a 13.5% reduction in the effective Newton's constant governing linear growth.

We implement this parameterization in MGCAMB and perform a suite of twelve Markov Chain Monte Carlo analyses using Cobaya with Planck 2018 (TT, TE, EE + lensing) likelihoods, combined individually with SDSS redshift-space distortion (RSD), baryon acoustic oscillation (BAO), and Pantheon+ Type Ia supernova datasets. When growth-rate data are included (Planck + RSD), the fixed prediction remains statistically indistinguishable from ΛCDM ($\Delta\chi^2 = +1.34$), with all standard cosmological parameters unchanged within their uncertainties. When the modification amplitude is allowed to float, the Planck + RSD posterior yields $\mu_0 = +0.024 \pm 0.123$, placing the predicted value within $1.3\,\sigma$ of the best fit. The model produces a downward shift in σ_8 from 0.813 to 0.800 without degrading the CMB fit. As expected from the $\Sigma = 1$ constraint, BAO angular positions and supernova luminosity distances are unaffected. We present forecasts indicating that Euclid and DESI will achieve the sensitivity required to detect or exclude a modification at this amplitude. All chains, configuration files, and analysis scripts are publicly available.

1 Introduction

The ΛCDM model provides an excellent fit to cosmic microwave background (CMB) anisotropies [Planck Collaboration, 2020b] and large-scale structure observations. Nevertheless, persistent discrepancies between early- and late-time determinations of the matter clustering amplitude—typically characterized by σ_8 and $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$—have motivated investigations of controlled late-time modifications to cosmic growth that preserve early-universe consistency [Abbott et al., 2022, Heymans et al., 2021, Li et al., 2023]. Weak lensing surveys report σ_8 values lower than the Planck ΛCDM prediction of $\sigma_8 \approx 0.81$ [Planck Collaboration, 2020b], a discrepancy at the 2–3σ level that has persisted across independent experiments and analysis pipelines.

A general phenomenological approach encodes deviations from General Relativity (GR) through the μ–Σ parameterization [Bean & Tangmatitham, 2010, Daniel et al., 2010, Pogosian & Silvestri, 2016], in which $\mu(a)$ modifies the Poisson equation governing matter clustering and $\Sigma(a)$ modifies the lensing potential. This framework is widely adopted by major surveys including DES [Abbott et al., 2022], KiDS [Heymans et al., 2021], and the upcoming Euclid mission [Frusciante et al., 2025]. Maintaining $\Sigma(a) = 1$ preserves photon geodesics and the CMB acoustic structure while allowing modifications to growth.

In this work, we examine a specific realization within this framework: a scale-independent, late-time suppression with $\mu(a) < 1$. The functional form is

$$\mu(a) = \frac{H^2_{\Lambda\text{CDM}}(a)}{H^2_{\Lambda\text{CDM}}(a) + \beta\,\mathcal{E}(a)}, \tag{1}$$

where

$$\mathcal{E}(a) = \exp(1 - 1/a) \tag{2}$$

is an activation function that vanishes as $a \to 0$ and reaches unity at $a = 1$. This ensures negligible modification during radiation domination and recombination. For the coupling $\beta = \Omega_m/2 \approx 0.1575$ (with $\Omega_m = 0.315$), the present-day value is

$$\mu(z{=}0) = \frac{1}{1+\beta} \approx 0.865, \tag{3}$$

corresponding to a 13.5% suppression of the effective gravitational coupling for matter perturbations.

The functional form of $\mathcal{E}(a)$ is motivated by the hypothesis that late-time structure formation introduces an effective informational entropy contribution to the matter sector. Specifically, the activation function arises in a companion analysis [Mahaffey, 2026] from integrating the cumulative decoherence rate weighted by cosmic horizon thermodynamics (Bekenstein-Hawking entropy, Gibbons-Hawking temperature, Landauer's principle): the information surface density on the apparent horizon scales as $1/a^2$ during matter domination, and its integral yields $\ln \mathcal{E} = 1 - 1/a$. The coupling $\beta = \Omega_m/2$ follows from the virial theorem applied to the partition of gravitational energy between geometric curvature and informational entropy. The rational form of Eq. (1) then follows directly from adding the informational entropy to the Cai-Kim horizon first law [Mahaffey, 2026], rather than being fitted to data. Independently of the theoretical motivation, the rational form represents the minimal single-parameter realization among monotonic activation functions that vanish at early times and saturate at $a = 1$: it preserves $\mu \to 1$ at high redshift by construction while fixing $\mu(z{=}0)$ through the single amplitude β. This work focuses on the phenomenological implications within the standard μ–Σ framework; no specific covariant action is assumed. The model is implemented using the publicly available MGCAMB code [Wang et al., 2023], which provides native μ–Σ support within the CAMB Boltzmann solver [Lewis et al., 2000].

The signature $\mu < 1$ with $\Sigma = 1$ has received comparatively less attention in the literature than $\mu > 1$ scenarios. Most modified gravity theories—including $f(R)$ gravity, DGP braneworld models, and scalar-tensor theories—predict $\mu > 1$, i.e., *enhanced* growth [Pogosian & Silvestri, 2016]. A suppression ($\mu < 1$) shifts σ_8 downward, in contrast to most modified gravity scenarios in the current literature, which predict $\mu \geq 1$. This places the model in a less explored region of the μ–Σ parameter space relative to most modified gravity scenarios in the current literature.

This paper is organized as follows. Section 2 defines the model and its implementation. Section 3 describes the datasets and methodology. Section 4 presents results from the full suite of MCMC analyses. Section 5 discusses the implications, including forecasts for upcoming surveys. Section 6 concludes.

2 Model Definition and Implementation

2.1 Modified Poisson Equations

In the μ–Σ framework, the modified Poisson equations in Fourier space are

$$k^2\Phi = -4\pi G\,\mu(a)\,a^2\,\bar{\rho}_m\,\delta_m, \tag{4}$$
$$k^2(\Phi+\Psi) = -8\pi G\,\Sigma(a)\,a^2\,\bar{\rho}_m\,\delta_m, \tag{5}$$

where Φ and Ψ are the scalar metric potentials in the conformal Newtonian gauge, $\bar{\rho}_m$ is the mean matter density, and δ_m is the matter density contrast. The function $\mu(a)$ modifies the growth of density perturbations, while $\Sigma(a)$ modifies the lensing potential $(\Phi+\Psi)/2$. General Relativity corresponds to $\mu = \Sigma = 1$.

We impose $\Sigma(a) = 1$ identically, ensuring that photon geodesics, gravitational lensing, and the CMB acoustic structure remain standard. We verify explicitly that the CMB lensing power spectrum changes by less than 0.5% relative to ΛCDM, and that the ISW contribution at $\ell < 30$ remains well within cosmic variance. The function $\mu(a)$ is given by Eq. (1).

2.2 MGCAMB Implementation

We implement the model using MGCAMB v1.5.2 [Wang et al., 2023], which provides a native μ–Σ parameterization within CAMB. MGCAMB's `pure_MG` mode implements the time dependence

$$\mu(a) = 1 + \mu_0\,\Omega_{\rm DE}(a), \qquad \Sigma(a) = 1 + \Sigma_0\,\Omega_{\rm DE}(a), \tag{6}$$

where $\Omega_{\rm DE}(a) = \Omega_\Lambda/[\Omega_m\,a^{-3} + \Omega_\Lambda]$ is the dark energy density fraction (neglecting radiation at late times). Here μ_0 denotes the *deviation* from GR at $z = 0$, such that the present-day value of μ is $\mu(z{=}0) = 1 + \mu_0$.

To match our model at $z = 0$, we set $\mu_0 = -0.135$ and $\Sigma_0 = 0$. The MGCAMB parameterization in Eq. (6) approximates the exact form in Eq. (1) to within 1–2.5% at intermediate redshifts ($0.5 \lesssim z \lesssim 2$), with exact agreement at $z = 0$ and $z \gg 1$. The deviation arises because $\mathcal{E}(a)$ and $\Omega_{\rm DE}(a)$ have slightly different transition profiles; however, both vanish at early times and approach their maximum at $z = 0$.

Notation convention. Throughout this paper, μ_0 refers to the MGCAMB parameter $\mu(z{=}0) - 1$, the deviation from GR. A negative value ($\mu_0 < 0$) corresponds to growth suppression ($\mu < 1$). The predicted value is $\mu_0 = -0.135$.

The MGCAMB configuration parameters used in all runs are: `MG_flag` = 1, `pure_MG_flag` = 2, `musigma_par` = 1, `GRtrans` = 0.001 (the scale factor below which GR is enforced), and `sigma0` = 0.

3 Data and Methodology

3.1 Datasets

We employ the following datasets:

Planck 2018 CMB. We use the full Planck 2018 likelihood suite [Planck Collaboration, 2020a,b]: low-ℓ temperature and polarization (`lowl.TT`, `lowl.EE`), high-ℓ temperature and polarization (`plik_lite TTTEEE`), and CMB lensing (`lensing.CMBMarged`).

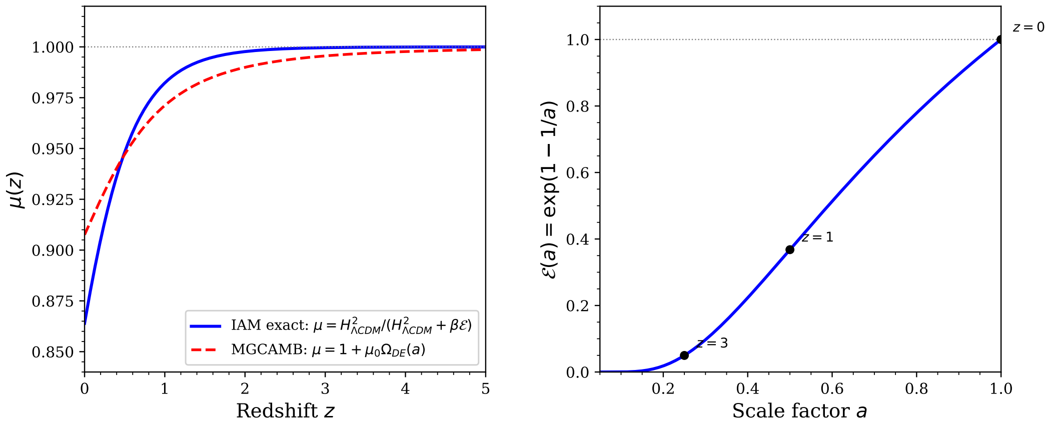

Figure 1: Left: Gravitational coupling $\mu(z)$ for the exact form (Eq. 1, solid blue) and the MGCAMB approximation (Eq. 6, dashed red). The two agree at $z = 0$ and $z \gg 1$, with 1–2.5% deviation at intermediate redshifts. Right: Activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, showing negligible modification at $z > 3$ and full activation at $z = 0$.

Redshift-space distortions (RSD). Growth rate measurements $f\sigma_8(z)$ from the SDSS-III BOSS DR12 [Alam et al., 2017] and SDSS-IV eBOSS DR16 [Alam et al., 2021] galaxy surveys. These measurements directly probe the combination of the growth rate and clustering amplitude that $\mu(a)$ modifies.

Baryon acoustic oscillations (BAO). BAO angular scale measurements from SDSS DR12 [Alam et al., 2017], eBOSS DR16 LRG, QSO, and Lyman-α samples [Alam et al., 2021]. A key point of the sector assignment: although BAO features are imprinted in the matter distribution, their *angular positions* are measured via photons traveling on null geodesics. Since $\Sigma = 1$, the angular diameter distance $D_A(z)$ and hence the BAO angular scale $\theta_{\rm BAO}$ are identical to ΛCDM. The amplitude of the BAO feature may be affected by the modified growth rate, but the geometric position—which is the primary BAO observable—is not.

Type Ia supernovae. The Pantheon+ sample of 1701 light curves in the redshift range $0.001 < z < 2.26$ [Brout et al., 2022]. As photon-sector distance measurements with $\Sigma = 1$, supernova luminosity distances are predicted to be identical to ΛCDM at the perturbation level considered here.

3.2 Analysis Strategy

We perform twelve MCMC analyses organized into four dataset combinations, each with three model configurations:

1. **Fixed** μ_0: $\mu_0 = -0.135$ (the predicted value, with zero additional free parameters relative to ΛCDM).
2. **Floating** μ_0: μ_0 sampled freely with a flat prior $\mu_0 \in [-0.5, +0.2]$ (one additional free parameter).
3. **ΛCDM baseline**: $\mu_0 = 0$, $\Sigma_0 = 0$ (standard GR, identical MGCAMB infrastructure).

The four dataset combinations are:

- Planck only (Runs A, B, C),
- Planck + RSD (Runs D, E, F),
- Planck + BAO (Runs G, H, I),
- Planck + Pantheon+ (Runs J, K, L).

The six standard ΛCDM parameters are sampled in all runs: $\{\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s\}$. Derived parameters include σ_8, Ω_Λ, and S_8.

All chains are run with the Cobaya MCMC sampler [Torrado & Lewis, 2021] using a Gelman–Rubin convergence criterion $R - 1 < 0.01$ for both means and bounds, with a minimum of 20,000 accepted samples after burn-in. The ΛCDM baseline runs use identical MGCAMB infrastructure (with $\mu_0 = 0$, $\Sigma_0 = 0$) to ensure that any differences arise solely from the modified gravity parameter rather than from code-level systematics.

4 Results

4.1 Planck Only (Runs A, B, C)

Table 1 presents the marginalized parameter constraints from the Planck-only analysis. The fixed prediction $\mu_0 = -0.135$ yields $\Delta\chi^2 = +1.43$ relative to ΛCDM. Given that the fixed model has the same number of free parameters as ΛCDM, this difference is statistically negligible ($p = 0.23$ for $\Delta\chi^2 = 1.43$ with zero additional parameters).

When μ_0 is allowed to float (Run B), the posterior peaks near $\mu_0 = 0.006 \pm 0.156$, consistent with GR but with broad uncertainties that accommodate the predicted value at 0.9σ. The broad posterior reflects the limited sensitivity of the CMB alone to late-time growth modifications when $\Sigma = 1$.

Table 1: Planck-only parameter constraints (Runs A, B, C). Values are posterior means $\pm 1\sigma$. $\Delta\chi^2$ is relative to the ΛCDM baseline (Run C).

Parameter	ΛCDM (C)	Fixed μ_0 (A)	Floating μ_0 (B)
H_0 [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	67.19 ± 0.54	67.06 ± 0.51	67.16 ± 0.51
σ_8	0.8139 ± 0.0062	0.8014 ± 0.0057	0.8146 ± 0.0157
$\Omega_b h^2$	0.02236 ± 0.00014	0.02234 ± 0.00014	0.02236 ± 0.00014
$\Omega_c h^2$	0.1202 ± 0.0012	0.1204 ± 0.0012	0.1202 ± 0.0013
τ	0.054 ± 0.007	0.056 ± 0.007	0.055 ± 0.008
n_s	0.9645 ± 0.0041	0.9639 ± 0.0040	0.9645 ± 0.0042
$\ln(10^{10} A_s)$	3.044 ± 0.014	3.049 ± 0.014	3.046 ± 0.016
μ_0	0 (fixed)	-0.135 (fixed)	0.006 ± 0.156
$\chi^2_{\rm min}$	983.14	984.57	981.24
$\Delta\chi^2$	—	$+1.43$	-1.90

4.2 Planck + RSD (Runs D, E, F)

The combination of Planck with growth rate data provides a direct test of the gravitational coupling modification, since $f\sigma_8(z)$ measurements are sensitive to $\mu(a)$. Table 2 presents the results.

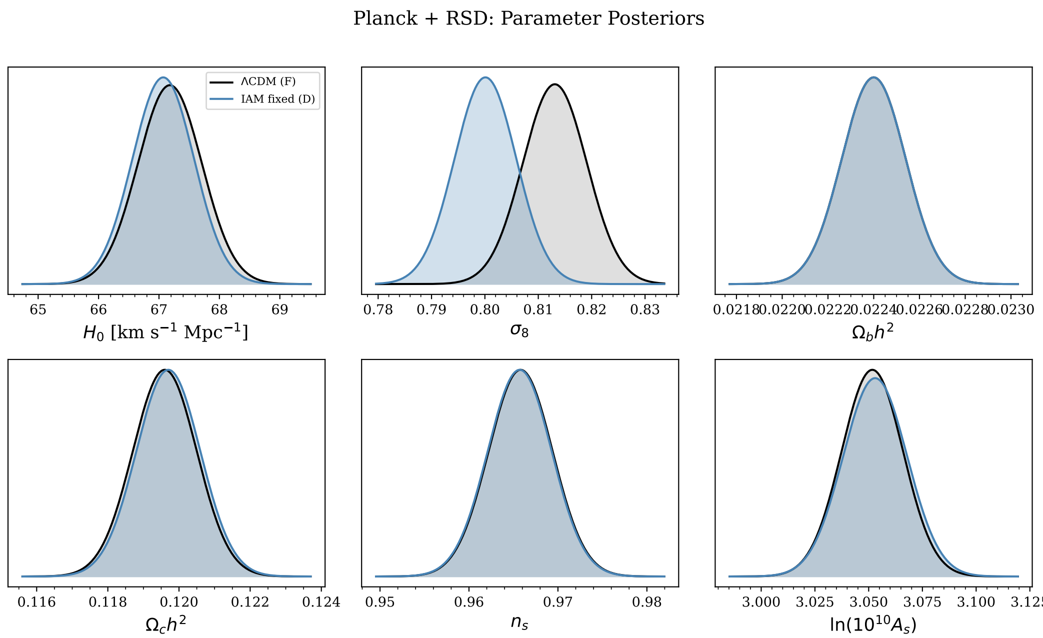

Figure 2: Marginalized posterior distributions for cosmological parameters from Planck + RSD: ΛCDM baseline (gray, Run F) and fixed $\mu_0 = -0.135$ (blue, Run D). The σ_8 posterior is shifted to lower values; other parameters are unchanged.

The fixed prediction ($\mu_0 = -0.135$) achieves $\Delta\chi^2 = +1.34$ relative to ΛCDM, a reduction from the Planck-only value of +1.43. This indicates that growth rate data do not penalize the growth suppression.

The principal parameter shift is in σ_8: the model yields $\sigma_8 = 0.8001 \pm 0.0061$ compared to 0.8131 ± 0.0059 for ΛCDM, a reduction of 0.013. All other cosmological parameters remain unchanged within their uncertainties, confirming that the μ modification acts selectively on growth without introducing parameter degeneracies.

When μ_0 is allowed to float (Run E), the posterior yields $\mu_0 = +0.024 \pm 0.123$, with the predicted value lying $1.3\,\sigma$ from the best fit.

Table 2: Planck + RSD parameter constraints (Runs D, E, F). Values are posterior means $\pm 1\sigma$.

Parameter	ΛCDM (F)	Fixed μ_0 (D)	Floating μ_0 (E)
H_0 [km s^{-1} Mpc^{-1}]	67.50 ± 0.41	67.47 ± 0.42	67.10 ± 0.52
σ_8	0.8131 ± 0.0059	0.8001 ± 0.0061	0.8147 ± 0.0129
$\Omega_b h^2$	0.02240 ± 0.0001	0.02240 ± 0.0001	0.02240 ± 0.0001
$\Omega_c h^2$	0.1196 ± 0.0009	0.1197 ± 0.0009	0.1196 ± 0.0009
Ω_Λ	0.6867 ± 0.0056	0.6864 ± 0.0057	0.6867 ± 0.0056
n_s	0.9658 ± 0.0036	0.9657 ± 0.0036	0.9658 ± 0.0036
$\ln(10^{10}A_s)$	3.0517 ± 0.0143	3.0530 ± 0.0149	3.0520 ± 0.0150
μ_0	0 (fixed)	-0.135 (fixed)	$+0.024 \pm 0.123$
$\chi^2_{\rm min}$	993.86	995.20	989.87
$\Delta\chi^2$	—	+1.34	−3.99

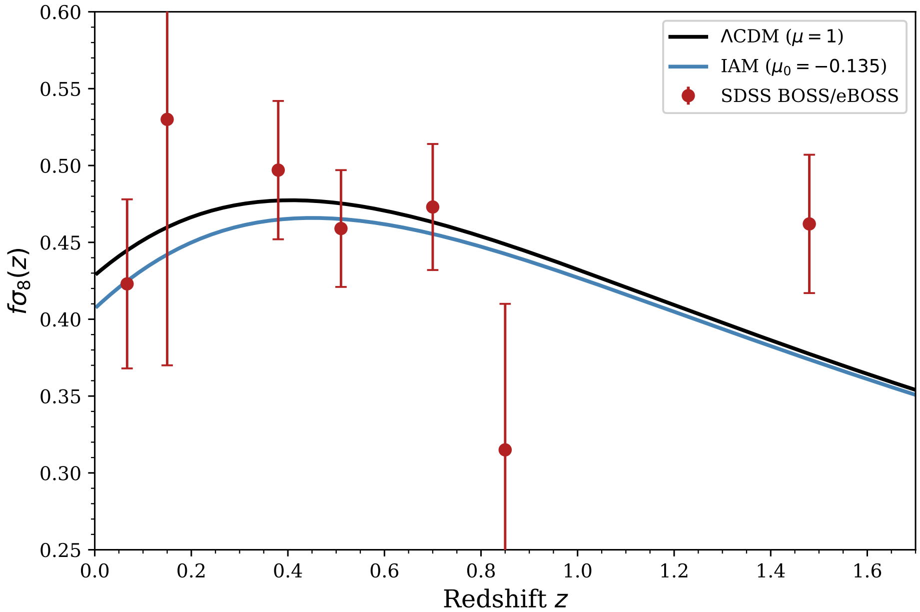

Figure 3: Growth rate $f\sigma_8(z)$ for ΛCDM (black) and fixed $\mu_0 = -0.135$ (blue), compared to SDSS BOSS/eBOSS measurements (red points with 1σ error bars). The $\mu < 1$ model predicts systematically lower $f\sigma_8$ at all redshifts.

4.3 Planck + BAO (Runs G, H, I)

BAO measurements constrain the angular diameter distance $D_A(z)$ and the Hubble parameter $H(z)$ through the transverse and line-of-sight baryon acoustic scales. Since these are photon-sector observables measured via null geodesics, and $\Sigma = 1$ in our model, the BAO angular scale is predicted to be identical to ΛCDM at the perturbation level.

Table 3 presents the results.

Table 3: Planck + BAO parameter constraints (Runs G, H, I). Values are posterior means $\pm 1\sigma$.

Parameter	ΛCDM (I)	Fixed μ_0 (G)	Floating μ_0 (H)
H_0 [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	67.56 ± 0.45	67.51 ± 0.43	67.56 ± 0.46
σ_8	0.8115 ± 0.0059	0.7981 ± 0.0063	0.8117 ± 0.0164
μ_0	0 (fixed)	-0.135 (fixed)	$+0.002 \pm 0.158$
$\chi^2_{\rm min}$	1017.71	1020.03	1017.90
$\Delta\chi^2$	—	$+2.32$	$+0.19$

4.4 Planck + Pantheon+ (Runs J, K, L)

Pantheon+ supernovae measure luminosity distances, which are again photon-sector observables with $\Sigma = 1$. Table 4 presents the results.

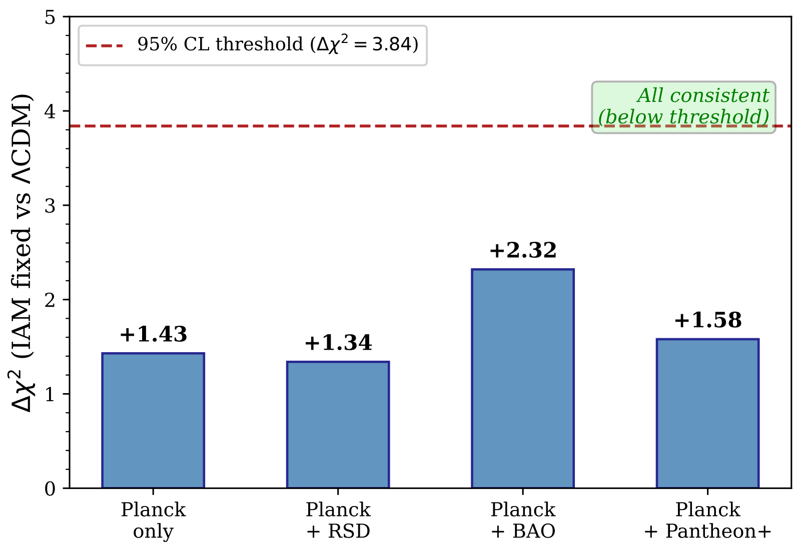

Figure 4: $\Delta\chi^2$ for the fixed $\mu_0 = -0.135$ model relative to ΛCDM across all dataset combinations. The dashed line indicates $\Delta\chi^2 = 3.84$ (95% confidence level threshold for one parameter). All values are below this threshold.

Table 4: Planck + Pantheon+ parameter constraints (Runs J, K, L). Values are posterior means $\pm 1\sigma$.

Parameter	ΛCDM (L)	Fixed μ_0 (J)	Floating μ_0 (K)
H_0 [km s^{-1} Mpc^{-1}]	67.11 ± 0.51	67.03 ± 0.52	67.06 ± 0.52
σ_8	0.8129 ± 0.0060	0.8000 ± 0.0057	0.8124 ± 0.0165
μ_0	0 (fixed)	-0.135 (fixed)	-0.005 ± 0.162
$\chi^2_{\rm min}$	2416.00	2417.58	2415.40
$\Delta\chi^2$	—	+1.58	−0.60

4.5 Combined Summary

Table 5 summarizes the $\Delta\chi^2$ values and σ_8 shifts across all dataset combinations.

Table 5: Summary of $\Delta\chi^2$ (fixed $\mu_0 = -0.135$ relative to ΛCDM) and σ_8 shifts across all dataset combinations. $\Delta\sigma_8 = \sigma_8(\text{IAM}) - \sigma_8(\Lambda\text{CDM})$.

Dataset	$\Delta\chi^2$ (fixed)	σ_8 (ΛCDM)	σ_8 (fixed μ_0)
Planck only	+1.43	0.8139	0.8014
Planck + RSD	+1.34	0.8131	0.8001
Planck + BAO	+2.32	0.8115	0.7981
Planck + Pantheon+	+1.58	0.8129	0.8000

5 Discussion

5.1 Compatibility with Current Data

The principal result of this analysis is that a zero-parameter theoretical prediction—a 13.5% suppression of the effective gravitational coupling at $z = 0$, derived from $\beta = \Omega_m/2$—survives the full Planck 2018 likelihood without tension. The $\Delta\chi^2$ values of +1.43 (Planck only) and +1.34 (Planck + RSD) correspond to less than 1σ for a model with the same number of free parameters. Most proposed late-time modifications to gravity are excluded or severely constrained by Planck; the compatibility of this prediction with current data is not guaranteed *a priori* and constitutes a meaningful consistency check.

The result is stable across dataset combinations. Adding growth rate data—which directly probe the modification—does not increase $\Delta\chi^2$, indicating that the predicted growth suppression is compatible with observed $f\sigma_8(z)$ measurements.

5.2 The σ_8 Shift

The model shifts σ_8 downward from ~ 0.813 to ~ 0.800 across all dataset combinations. This 1.6% reduction is a direct consequence of $\mu < 1$: the weakened gravitational source term in the Poisson equation reduces the growth rate and hence σ_8, while leaving the CMB primary anisotropy spectrum unchanged (since $\Sigma = 1$). The magnitude of the shift is determined entirely by the predicted $\mu_0 = -0.135$; it is not fitted to any late-time clustering data. We emphasize that this shift is a zero-parameter prediction: $\beta = \Omega_m/2$ is derived, not adjusted, and the direction of the shift—toward the values reported by weak lensing surveys—was not used as input.

For comparison, KiDS-1000 reports $S_8 = 0.759^{+0.024}_{-0.021}$ [Heymans et al., 2021] and DES Y3 reports $S_8 = 0.776 \pm 0.017$ [Abbott et al., 2022], both lower than the Planck ΛCDM value. Whether the shift produced by $\mu_0 = -0.135$ quantitatively reduces the discrepancy between CMB and weak lensing constraints requires a joint analysis with KiDS and DES likelihoods, which is beyond the scope of this work.

5.3 Comparison with Other Modified Gravity Models

The combination $\mu < 1$ and $\Sigma = 1$ has received comparatively less observational attention than $\mu > 1$ scenarios. Most well-studied theories predict $\mu > 1$: $f(R)$ gravity, DGP braneworld models, and Horndeski scalar-tensor theories generically produce $\mu \geq 1$ [Pogosian & Silvestri, 2016]. The suppression $\mu < 1$ with Σ unmodified is observationally distinguishable from these models.

Current constraints are already informative. DES Y3 reports $\mu_0 = -0.4 \pm 0.4$ [Abbott et al., 2023], Andrade et al. (2024) find $\mu_0 - 1 = 0.02 \pm 0.19$ from ACT+WMAP+SDSS [Andrade et al., 2024], and DESI DR1 full-shape analysis yields $\mu_0 = 0.11^{+0.45}_{-0.54}$ [DESI Collaboration, 2025]. The predicted value $\mu_0 = -0.135$ lies within the allowed region of all current constraints.

5.4 Forecasts for Upcoming Surveys

Next-generation surveys will substantially improve sensitivity to μ_0. Fisher forecasts based on published survey specifications yield the projected constraints shown in Table 6.

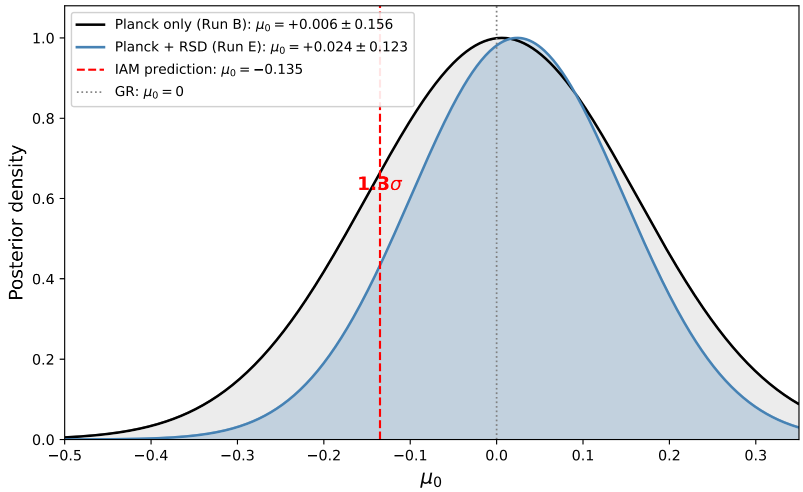

Figure 5: Marginalized posterior for μ_0 from Planck only (Run B, gray) and Planck + RSD (Run E, blue). The vertical dashed line indicates the predicted value $\mu_0 = -0.135$. Both posteriors are consistent with the prediction.

Table 6: Projected Fisher forecast constraints on $\mu_0 = -0.135$ with $\Sigma = 1$. Current constraints are from this work (Planck + RSD). Projected sensitivities are based on published survey specifications and assume $\Sigma = 1$ as a prior.

Survey / Configuration	$\sigma(\mu_0)$	$\lvert\mu_0\rvert/\sigma(\mu_0)$
Planck + RSD (this work)	~ 0.12	~ 1.1
Euclid (pessimistic)	~ 0.06	~ 2.2
Euclid (optimistic)	~ 0.04	~ 3.4
DESI Year 5 ($f\sigma_8$ only)	~ 0.10	~ 1.4
Euclid + DESI Y5 combined	~ 0.025	~ 5.4

Euclid alone is projected to reach $\sigma(\mu_0) \approx 0.04$ in its optimistic configuration, which would test the predicted amplitude at the $\sim 3\sigma$ level. The combination of Euclid and DESI Year 5 data would further tighten the constraint to $\sigma(\mu_0) \approx 0.025$. These projections assume $\Sigma = 1$ is imposed as a prior; allowing both μ_0 and Σ_0 to float would weaken the constraints.

5.5 Limitations and Future Directions

Several limitations should be noted:

1. **Phenomenological framework.** This analysis tests a μ–Σ parameterization only. The activation function $\mathcal{E}(a)$ and coupling $\beta = \Omega_m/2$ are derived from horizon thermodynamics and the virial theorem in a companion paper [Mahaffey, 2026], but no covariant action is assumed in the present work. The results stand independently of the theoretical motivation: any model predicting $\mu_0 \approx -0.135$ with $\Sigma = 1$ would face the same observational tests.

2. **Perturbation-level only.** The background cosmology (Friedmann equation) is standard ΛCDM throughout. Extension to a background-level modification is deferred to future work.

3. **MGCAMB approximation.** The MGCAMB parameterization $\mu(a) = 1 + \mu_0\,\Omega_{\rm DE}(a)$ approximates but does not exactly reproduce the target form in Eq. (1). The 1–2.5% deviation at intermediate redshifts is subdominant to current observational uncertainties but may become relevant at Euclid-level precision.

4. **Linear scales only.** Nonlinear structure formation (cluster counts, small-scale lensing) is not included. The μ modification applies to linear perturbation theory; its extension to the nonlinear regime requires N-body simulations or validated fitting functions.

5. **Weak lensing likelihoods.** The σ_8 shift is compared to the reported discrepancy between CMB and weak lensing constraints but is not tested directly against KiDS, DES, or HSC likelihoods. A joint analysis is planned for future work.

6 Conclusions

We have performed a comprehensive perturbation-level validation of a scale-independent, late-time growth suppression parameterized as $\mu_0 = -0.135$ within the standard μ–Σ framework. Our principal findings are:

1. A fixed 13.5% suppression of the gravitational coupling at $z = 0$ is statistically indistinguishable from ΛCDM against Planck 2018 CMB data ($\Delta\chi^2 = +1.43$) and remains so when combined with SDSS growth rate data ($\Delta\chi^2 = +1.34$).

2. The model shifts σ_8 from ~ 0.813 to ~ 0.800 without degrading any other cosmological parameter.

3. The prediction lies within $1.3\,\sigma$ of the best-fit μ_0 when the parameter is allowed to float against Planck + RSD data.

4. BAO and supernova datasets, which are photon-sector observables, are unaffected by the perturbation-level modification, as expected from $\Sigma = 1$.

5. Euclid and DESI are projected to reach the sensitivity required to detect or exclude a modification at this amplitude.

The parameterization $\mu < 1$, $\Sigma = 1$ occupies a less explored region of the modified gravity parameter space relative to $f(R)$, DGP, and most scalar-tensor theories. Current data do not disfavor this parameterization, and it produces specific predictions testable by next-generation surveys. The theoretical derivation of the functional form and coupling constant is presented in a companion paper [Mahaffey, 2026].

Acknowledgments

The author thanks the developers of MGCAMB [Wang et al., 2023], Cobaya [Torrado & Lewis, 2021], and CAMB [Lewis et al., 2000] for making their codes publicly available. Computational resources were provided by personal hardware. This research made use of the Planck Legacy Archive.

Data Availability

All MCMC chains, Cobaya YAML configuration files, convergence diagnostics, analysis scripts, and this manuscript are publicly available at:

`https://github.com/hmahaffeyges/IAM-Validation`

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